THE QUASI-FINITE LOCUS IS OPEN AND COMMUTES WITH BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is one paragraph reducing all four assertions to the scheme case through a single double presentation; nothing is left to the reader beyond applying the cited scheme statements.

Editorial note. The six references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Local quasi-finiteness is etale local on the source-and-target. Descent, Remark descent-remark-list-local-source-target, used at proof lines 5363-5365. Lets the statements be checked on the scheme presentations U -> V and U' -> V'.
- Pointwise and global forms of the source-and-target locality for this chapter. Lemmas lemma-local-source-target-at-point and lemma-local-source-target-global-implies-local, used at proof lines 5366-5370. Convert the local-on-presentations statements back to f and f'.
- Openness and base change behaviour of the quasi-finite locus for schemes. Morphisms, Lemmas morphisms-lemma-base-change-quasi-finite and morphisms-lemma-quasi-finite-points-open, used at proof lines 5371-5373. Are the scheme inputs for parts (1), (2) and (3).
- Quasi-finiteness is local quasi-finiteness plus quasi-compactness, which is stable under base change. Lemma lemma-base-change-quasi-compact, used at proof line 5374. Deduces part (4) from part (3).

Source reference. Original remark 04R3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6171–6171. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03HF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4565–4565. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 04NC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6073–6073. Navigation pointer into the Stacks

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Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 04RF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6183–6183. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 01TM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2379–2379. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 01TI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2375–2375. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ and $g : Y' \to Y$ be morphisms of algebraic spaces over S. Denote $f' : X' \to Y'$ the base change of f by g. Denote $g' : X' \to X$ the projection. Assume f is locally of finite type. Let $W \subset |X|$, resp. $W' \subset |X'|$ be the set of points where f, resp. f' is quasi-finite.*

1. *$W \subset |X|$ and $W' \subset |X'|$ are open,*
2. *$W' = (g')^{-1}(W)$, i.e., formation of the locus where f is quasi-finite commutes with base change,*
3. *the base change of a locally quasi-finite morphism is locally quasi-finite, and*
4. *the base change of a quasi-finite morphism is quasi-finite.*

Proof. Choose a scheme V and a surjective étale morphism $V \to Y$. Choose a scheme U and a surjective étale morphism $U \to V \times_Y X$. Choose a scheme V' and a surjective étale morphism $V' \to Y' \times_Y V$. Set $U' = V' \times_V U$ so that $U' \to X'$ is a surjective étale morphism as well. Picture

$$\begin{array}{ccc} U' & \longrightarrow & U \\ \downarrow & & \downarrow \\ V' & \longrightarrow & V \end{array} \quad \text{lying over} \quad \begin{array}{ccc} X' & \longrightarrow & X \\ \downarrow & & \downarrow \\ Y' & \longrightarrow & Y \end{array}$$

Choose $u \in |U|$ with image $x \in |X|$. The property of being "locally quasi-finite" is étale local on the source-and-target, see Descent, Remark 04R3. Hence Lemmas 04NC and 04RF apply and we see that $f : X \to Y$ is quasi-finite at x if and only if $U \to V$ is quasi-finite at u. Similarly for $f' : X' \to Y'$ and the morphism $U' \to V'$. Hence parts (1), (2), and (3) reduce to Morphisms, Lemmas 01TM and 01TI. Part (4) follows from (3) and Lemma 03HF. □